

A Visual Analysis of the Copenhagen Apartment Market

Exam Paper

Written Home Assignment

Course: Business Data Processing and Business Intelligence
Course code: CDSCV1005E

Student name: Torp, Aleksander
Student number: 158277

Date of submission: 18.12.23
Number of characters 17 955 characters
Number of pages: 10 pages

Table of Contents

1. Introduction.....	2
1.1. Problem Statement and Research Question.....	2
1.2. Delimitations	2
2. Methodology.....	3
2.1 Data Collection	3
2.1.1 Sales Prices	3
2.1.2 Apartment Location	4
2.1.3 Consumer Price Index.....	4
2.1.4 Area Demographics.....	4
2.2 Data Pre-processing	4
2.3 Data Storage.....	5
2.3.1 Preparation for Database	5
2.3.2 Choice of Database.....	7
3. The Dashboard	8
4. Findings.....	9
5. Conclusion	10
6. References.....	11
7. Appendices.....	12
7.1. Appendix A – List of Copenhagen City Areas Covered	12
7.2. Appendix B – SQL Code to Create Tables and Define Data Types	13
7.3. Appendix C – SQL Code to Adjust Prices for Inflation.....	14
7.4. Appendix D – Dashboard: Detailed Picture of Market Overview	15
7.5. Appendix E – Dashboard: Detailed Picture of Price Forecasting	16

1. Introduction

1.1. Problem Statement and Research Question

This study aims to provide a comprehensive visual analysis of the apartment market in Copenhagen, capturing the nuances of market trends and regional developments. The goal is to create an interactive dashboard that offers adjustable parameters for a wide range of users, combined with forecasting features to predict market trends in the near to medium term.

Obtaining an overview of housing markets can be complex. Many online platforms allow for individual property inquiries, but a consolidated view of the market is difficult to come by. This research addresses this gap by offering an efficient aggregated perspective of the owned apartment segment, a common and accessible housing option in Denmark. The insights derived from this study could benefit sellers, buyers, investors, construction firms, and policymakers in Denmark's capital.

While the outcome of this research can provide descriptive, visual analytics and predictive insights into one or several segments of Copenhagen's housing market for various stakeholders, this paper seeks to address the following research questions:

1. What is the price development for apartments in the range 80-100 square meters in Copenhagen the last 10 years?
2. What is the distribution in terms of price and units between the different city areas for such apartments today?

1.2. Delimitations

This investigation is deliberately narrowed to concentrate on a specific segment of the housing market, self-owned apartments. This focus is chosen to shed light on a significant portion of residential real estate that is both widespread and relevant to a diverse group of stakeholders, ranging from first-time homebuyers to investors. Nevertheless, this excludes other housing categories of relevance to many. Moreover, the geographical scope of the analysis is confined to many but not all city areas within central Copenhagen. It should be noted that the Greater Copenhagen region falls outside the scope of this study due to practical limitations in terms of data gathering. However, this exclusion allows for a more in-depth and concentrated analysis of the urban center of Denmark's capital.

2. Methodology

This section elaborates on the details on how necessary data was retrieved from various sources, the pre-processing performed, the characteristics of the data, and preparation of the data for storage and before proceeding to the analysis. Figure 1 outlines the overarching data flows and processes.

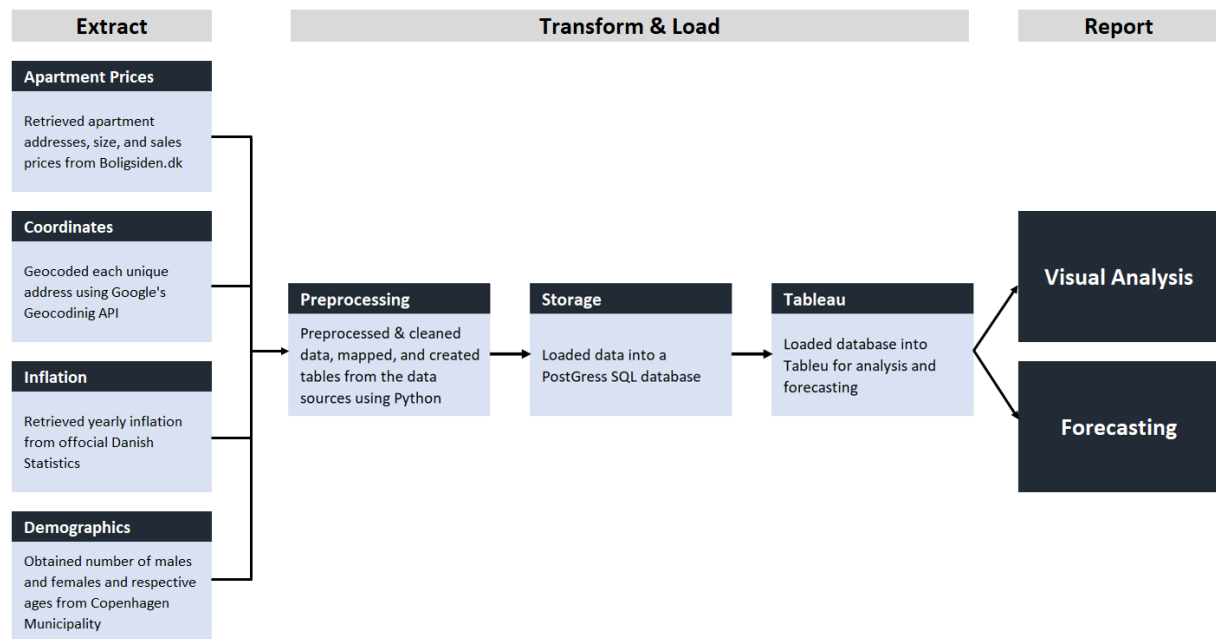


Figure 1: Process Diagram

2.1 Data Collection

In order to create the interactive dashboard and to perform analyses, information from multiple sources needed to be collected. This included downloading information from official sources, as well as retrieving data both by web scraping and Web API. As such, market data on apartments, respective coordinates, inflation for the time period and the latest demographic information was collected.

2.1.1 Sales Prices

No single source providing easy to access an overview of the housing market, or the necessary data to make one, was found online. Thus, the primary data was obtained through web scraping. The website Boligsiden.dk contains historical sales data on a variety of housing types across Copenhagen going back to 2000, providing a solid foundation for analysis. The data scraped was from the subcategory ownership apartments. Python and the library BeautifulSoup were used to fetch the data from the website. In total, 106 070 sales records were collected for the time period between 1st of January 2000 and 10th of August 2023 and stored in a CSV file. See Appendix A for the areas in Copenhagen covered.

2.1.2 Apartment Location

To perform geospatial analysis, the initial data scraped was inadequate since it only contained apartment locations in textual format, thus an apartment's address by itself was not enough. Conveniently, Google offers a Web API, Geocoding, for retrieving the coordinates of any given address (Google, n.d.). After creating a list of all the unique apartment addresses in the scraped dataset, their respective coordinates were retrieved by combining street address, house number, area name and zip code. In total, 51 173 coordinates were retrieved and stored in a CSV file.

2.1.3 Consumer Price Index

The initial sales data did not include sales prices adjusted for inflation. Therefore, the yearly consumer price index for the last twenty-three years in Copenhagen, with 2022 as the base year, was downloaded in a CSV file from Statistics Denmark (2023).

2.1.4 Area Demographics

Finally, demographic information, the number of males and females in different age brackets, was retrieved for the different regions in Copenhagen from the municipality's statistics bank in a CSV file (Københavns Kommune, 2023). The demographic information is available per quarter, and data from the third quarter of 2023 was downloaded to match the retrieval period of the sales data. However, the sales data also covers the whole area of the municipality Frederiksberg, and similar information on age distribution for both genders could not be retrieved for this area.

2.2 Data Pre-processing

After retrieval of all necessary data, the data sources were merged using Python and subject to quality control measures. To ensure reliability, ten separate random samples consisting of twenty to fifty sales records were extracted and manually compared with the records on Boligsiden. No mismatch was identified. Thereafter, 702 duplications were found and removed when checking against all columns and rows. Then, the data types for each variable were coerced to match the content, e.g., sales prices were set to float, year to integer, and so on. Next, no missing values were found. Finally, a visual inspection was performed to assess the distributions of values. From Figure 2, one sees what natural oscillations in terms of sales frequency between years and city areas may be. In contrast, the distribution of apartment sizes and sales prices exhibit strong skewness. Since, the data reflects the actual conditions of the housing market, no action were made before the analysis stage.

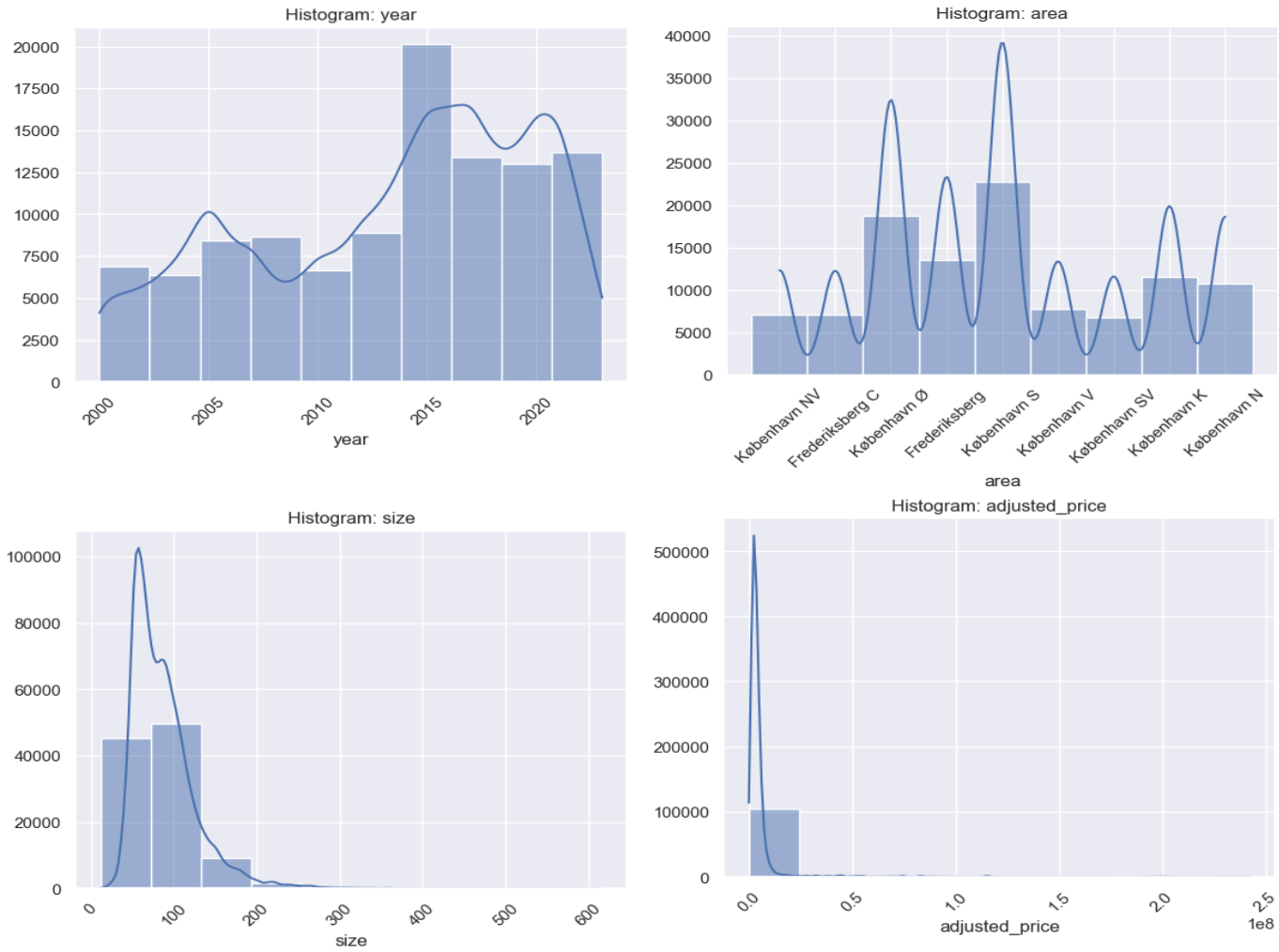


Figure 2: Distribution of Sales Year, Sales per Area, Apartment Size and Sales Price

2.3 Data Storage

The processes of retrieving the data and performing the necessary transformations before storing and analyzing the data involved multiple stages. Due to the nature of the data, the data collection and data preparation, Python with the Pandas library was primarily used since it offers the necessary flexibility in handling a broad range of tasks (scraping websites, usage of web APIs, complex data portioning, rearrangements and merging etc.) in the same environment. Moreover, the CSV storage format was used for both temporary saving between the different stages and uploading data to its final storage destination in a database. SQL was used to create the database, define tables, data types, and to populate the database, see Appendix B, following the data preparation with Python.

2.3.1 Preparation for Database

The initial dataset from Boligsiden consisted of the following elements: property addresses, transactional prices, apartment dimensions, sale dates, and a maximum of three sale records

per property unit. The first preparation step involved the generation of exclusive identifiers for each apartment using the address attribute, which is comprised of street name, house number, level, building location, postal code, and city area. Using Pandas, a unique list was extracted and stored in a separate CSV file and each apartment given separate apartment IDs. This was followed by the deletion of the 'placement within a building' attribute to mitigate potential privacy issues.

Next, the unique address list was used in conjunction with Google's Geocoding API to obtain respective coordinates. To ensure accuracy, five random samples consisting of twenty records were extracted to manually verify the coordinates fetched from Google against Google Maps. Afterward, all data were combined into one Dataframe, as shown in Figure 3.

apartment_id	area	zip_code	street_name	house_number	floor	size	price	latitude	longitude	date
1	Frederiksberg	2000	Ved Andebakken	3	2	59	3595000	55.6794	12.5281	2021-07-26
1	Frederiksberg	2000	Ved Andebakken	3	2	59	1875000	55.6794	12.5281	2012-12-17
1	Frederiksberg	2000	Ved Andebakken	3	2	59	1195000	55.6794	12.5281	2002-03-14
2	Frederiksberg C	1810	Platanvej	7	2	79	4545000	55.6729	12.5405	2021-07-26
2	Frederiksberg C	1810	Platanvej	7	2	79	1260000	55.6729	12.5405	2003-03-28

Figure 3: Combined Dataframe of Sales Data and Coordinates

This DataFrame was then segmented into seven discrete tables: sales, apartments, buildings, area names, zip code, street names. Then, numerical identifiers (for faster search) for each unique entry were generated and the tables stored in the CSV format. Conforming to the principles of the third normal form for databases by decomposing into multiple tables to avoid redundancy, partial or transitive dependencies, ensure functional dependency on primary keys, the partitioning process resulted in a snowflake schema configuration. Figure 4 shows the resulting relational tables, their interconnections, and the finalized database architecture.

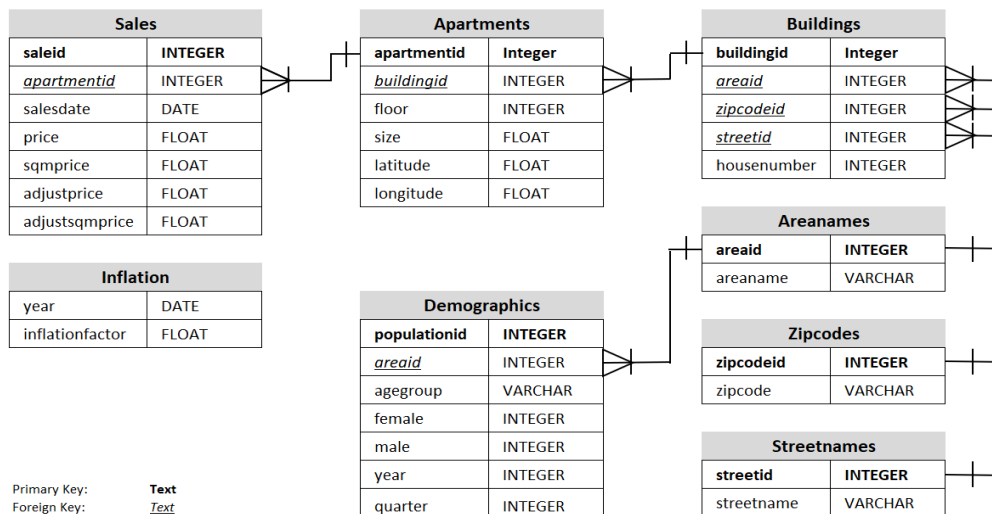


Figure 4: ERD Diagram of the Copenhagen Apartments Database

The sales table (weak entity) contains a unique primary key for each transaction and a foreign key to the related apartment, where one apartment has one or more sales. For each transaction, there is a sales date and price. When creating the table in Pandas, columns for the square meter price and the inflation adjusted price and square meter price were created and filled with zero. The inflation information was separated into an independent table to accommodate future updates, and SQL queries are used to computed adjusted prices for incoming data, see Appendix C for details.

The apartment table (weak entity) comprise information associated specifically with each apartment, such as which floor in the building it is in, its size (number of square meters), and the coordinates. Due to a one-to-one relationship between coordinates and apartments, the coordinates were included in this table, rather than being split into a separate table. Since multiple apartments may be in the same building, each apartment references a unique building in the buildings table.

The buildings table (weak entity) holds foreign keys to separate tables storing the areas, zip codes and street names of Copenhagen. These tables were necessary since one or more buildings can be in the same area, zip code and street. However, only one building can be associated with the above and a given house number.

The tables for area names, zip codes, and street names are all strong entities, all have primary keys referencing unique values within each respective field, without any foreign keys.

Finally, there is the demographics table, a weak entity, which has a many to one relationship with the area names table. Each entry specifies how many males or females in a certain age group, at a given quarter of the year, is associated with an area. Thus, one or multiple demographic records can be associated with an area.

2.3.2 Choice of Database

Since the data has clear relationships, e.g., between buildings and areas, a relational database system is suitable to effectively manage and analyze structured data. In this scenario the PostgreSQL database provide a robust framework to both preserve the structure of the data and facilitates seamless updates in the future. By scripting the database structure, see Appendix B, the data format and organization ensure alignment with the analytical requirements of this paper and efficient data retrieval.

Moreover, a relational database provides an advantage when combined with Tableau, the software subsequently used. By structuring data relationally, it enables utilizing Tableau's capabilities for creating the intended dynamic and interactive visualizations necessary to

inform the audience of this paper on the state of the apartment market, adjustable to individual information needs.

3. The Dashboard

After connecting the PostgreSQL database to Tableau, the multipurposed dashboard could be created. As illustrated in Figure 5, see Appendix D for more details, the dashboard consists of two pages the user can switch between by using the buttons in the upper left corner; the market overview page, which conveys an overview of the apartment market based on the user's selected parameters; and the price forecasting page, see appendix E for details, which falls outside the scope of this paper.

The user can set the following parameters: the price and size range of apartments, which city areas to include, whether to use price and/size filters at all, and whether to include the whole dataset or exclude outliers outside of the interquartile range. The last option is a measure to give users as much freedom as possible, while also addressing scenarios where the skewedness of the data, identified in section 2.2, could be an issue. By default, all filters are turned on.

Copenhagen Apartments

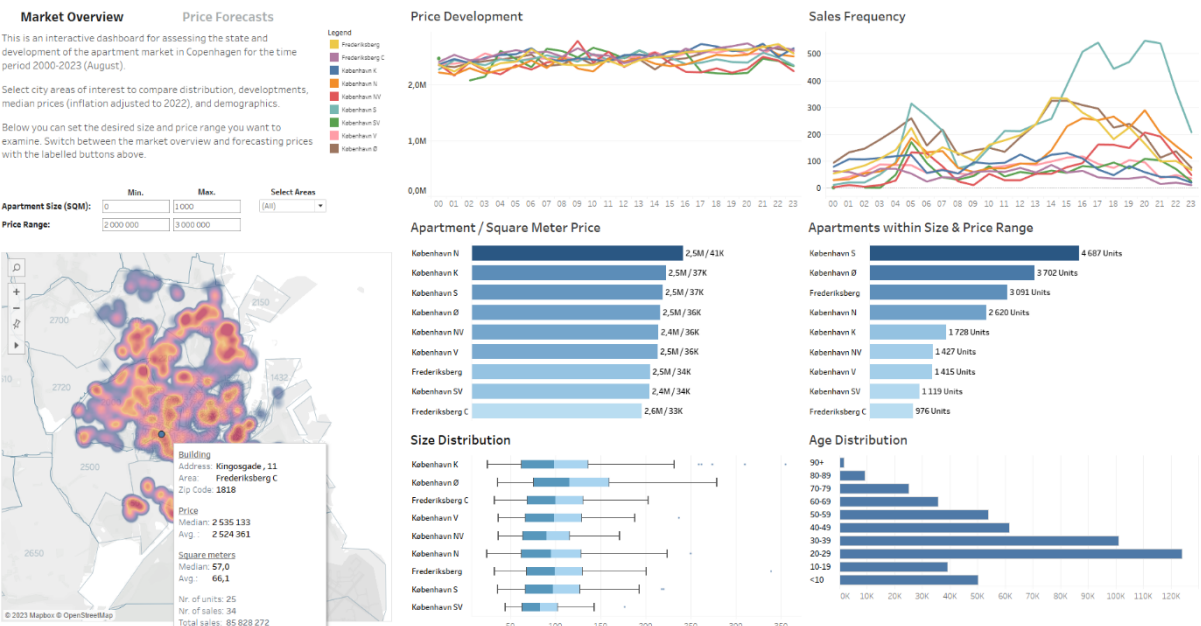


Figure 5: The Dashboard showing the Market Overview Page

As depicted in Figure 5, the dashboard utilizes seven visualizations to provide a holistic market assessment of the selected city areas, following the natural reading habit from left to right. The two first line plots convey the median price development per apartment and the frequency of apartment sales. The middle row, with bar graphs, shows the median inflation adjusted apartment and square meter price, and the number of units within the selected ranges. The bottom graphs display the distribution of apartment sizes through boxplots and the distribution

of the number of people within each age group. By hovering over any of the plots, additional details are revealed, such as the gender distribution in each age group. Finally, the heatmap in the lower left corner allows for geospatial analysis of the market segment and a visual interpretation of the apartment distribution. A more intense and brighter red color indicates relatively more expensive apartments in terms of median prices. By hovering over the datapoints, the user obtains building specific information concerning geographic location, median and average values on sales prices and apartment sizes, total number of units and sales volume sum for all units sold.

To preserve privacy rights of inhabitants, the prices are aggregated first for all sales per unique apartment and corresponding sales prices adjusted to 2022 prices, which is then aggregated to per building. For instance, as depicted in Figure 5/Appendix D, at Kingosgade 11, 1818, there is sales data for 25 unique apartments with a total of 34 sales between 2000 and 2023, and the median sales price is 2.5 million Danish Crowns with a median size of 57 square meters per unit.

4. Findings

To address both research questions of this paper, the size range was set to minimum 80 and maximum 100 square meters, the price range between 500 and 10 million DKK, and outlier filtering was turned on.

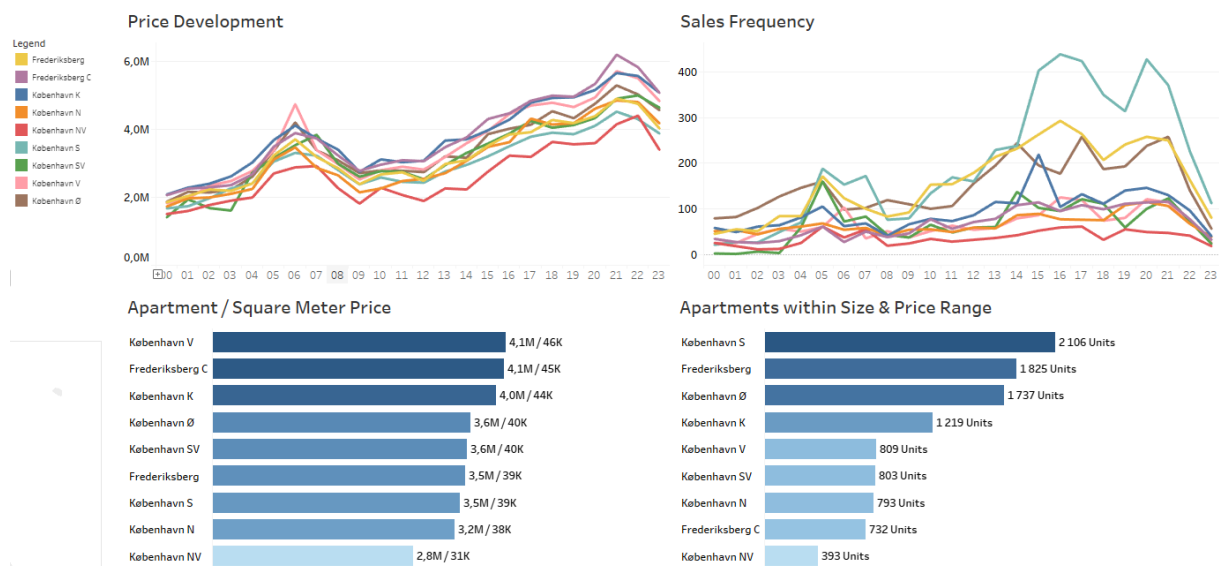


Figure 6: Market Overview for Apartments of Size 80-100 Square Meters in Copenhagen

Concerning the price development in the last decade, the upper left line graph in Figure 6 portrays a general positive price development for these apartments stretching back to year 2000, with a notable peak around 2006 and 2021-2022. The trend occurs across all city areas, and one notices a notable negative development from 2022 and onwards. This may relate to

delayed effects of monetary policies after the COVID-19 pandemic. Overall, there has been general trend of these apartments to increase in median selling price with around 1 million DKK in all areas. Copenhagen NV currently has the lowest median value at 3.4 million DKK in 2023, compared to 2.3 million DKK in 2013, and Frederiksberg C is at 5.1 million DKK in 2023, in contrast to 3.5 million DKK in 2013.

Regarding the price distribution between the different areas, the left bar chart in Figure 6 depicts that the most expensive apartments reside in Copenhagen V and K, in addition to Frederiksberg C, with the median price at 4 million DKK and upwards. At the other end of the spectrum one finds Copenhagen N and NV with median prices at 3.2 and 2.8 million DKK respectively. In terms of number of units within the 80-100 square meters range, one finds Copenhagen S at the top with more than 2100 units and Copenhagen NV at the bottom with less than 400 units. When examining the arrangement of the areas in terms of both median apartment values and units within the respective areas, no correlation is emerges.

5. Conclusion

This study has employed multiple data collection and analytical techniques to offer an in-depth visual analysis of Copenhagen's apartment market. After integrating various data sources, a multi-faceted database was constructed to facilitate the creation of an interactive dashboard. The dashboard, developed in Tableau, provides a dynamic platform to examine market trends and regional variations with adjustable parameters to cater to a wide spectrum of informational needs.

A visual analysis reveals a pronounced positive trajectory in the median selling prices of apartments sized 80-100 square meters across Copenhagen over the past decade, with distinct peaks in the pre-recession years as well as in 2021-2022. However, the market exhibits a downward trend from 2022, potentially a delayed consequence of monetary policy responses to the COVID-19 pandemic. Nonetheless, there has been a consistent increase of around 1 million DKK in median selling prices across all city areas, with Copenhagen NV at the lower end and Frederiksberg C at the higher end of the value spectrum. The visualizations also highlight the distribution of apartment prices and unit availability among various city areas, identifying Copenhagen V, K, and Frederiksberg C as the regions with the highest median apartment prices, while Copenhagen N and NV have options affordable to a broader audience. Disparities in pricing and availability aside, no clear correlation between median apartment values and the volume of units available emerged. The dashboard can provide both aggregated and detailed understandings of the owned apartment market dynamics in Copenhagen, and may serve as a resource for potential buyers and sellers, investors, construction companies, and policymakers,

6. References

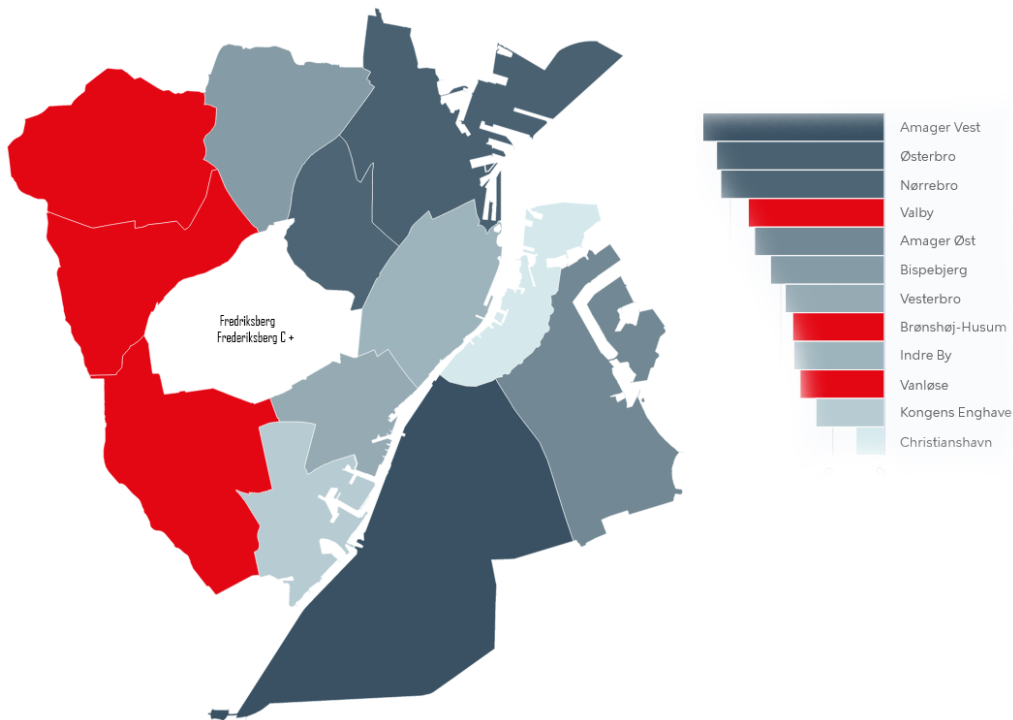
- Københavns Kommune. (2023, December). *Befolkning og fremskrivninger*. Retrieved from <https://kk.statistikbank.dk/statbank5a/default.asp?w=1536>
- Google. (n.d.). *Geocoding API Overview*. Retrieved from <https://developers.google.com/maps/documentation/geocoding/overview>
- Statistics Denmark (2023, November). *Consumer Price Index*. Retrieved from <https://www.dst.dk/en/Statistik/emner/oekonomi/prisindeks/forbrugerprisindeks>

7. Appendices

7.1. Appendix A – List of Copenhagen City Areas Covered

City Areas Abbreviations from Boligsiden

Frederiksberg	
Frederiksberg C	
Copenhagen K	Indre By + Christianshavn
Copenhagen NV	Bispebjerg
Copenhagen N	Nørrebro
Copenhagen Ø	Østerbro
Copenhagen SV	Kongens Enghave
Copenhagen S	Amager Vest + Øst
Copenhagen V	Vesterbro



7.2. Appendix B – SQL Code to Create Tables and Define Data Types

```
1  -- Creating the AreaNames Table
2  CREATE TABLE AreaNames (
3      AreaID INT PRIMARY KEY,
4      AreaName VARCHAR(255) NOT NULL
5  );
6
7  -- Creating the ZipCodes Table
8  CREATE TABLE ZipCodes (
9      ZipCodeID INT PRIMARY KEY,
10     ZipCode VARCHAR(10) NOT NULL
11 );
12 |
13 -- Creating the StreetNames Table
14 CREATE TABLE StreetNames (
15     StreetID INT PRIMARY KEY,
16     StreetName VARCHAR(255) NOT NULL
17 );
18
19 -- Creating the Buildings Table
20 CREATE TABLE Buildings (
21     BuildingID INT PRIMARY KEY,
22     AreaID INT,
23     ZipCodeID INT,
24     StreetID INT,
25     HouseNumber INT,
26     FOREIGN KEY (AreaID) REFERENCES AreaNames(AreaID),
27     FOREIGN KEY (ZipCodeID) REFERENCES ZipCodes(ZipCodeID),
28     FOREIGN KEY (StreetID) REFERENCES StreetNames(StreetID)
29 );
30
31 -- Creating the Apartments Table
32 CREATE TABLE Apartments (
33     ApartmentID INT PRIMARY KEY,
34     BuildingID INT,
35     Floor INT,
36     Size FLOAT,
37     Latitude FLOAT,
38     Longitude FLOAT,
39     FOREIGN KEY (BuildingID) REFERENCES Buildings(BuildingID)
40 );
41
42 -- Creating the Sales Table
43 CREATE TABLE Sales (
44     SaleID INT PRIMARY KEY,
45     ApartmentID INT,
46     SaleDate DATE,
47     Price FLOAT,
48     SqmPrice FLOAT,
49     AdjustedPrice FLOAT,
50     AdjustedSqmPrice FLOAT,
51     FOREIGN KEY (ApartmentID) REFERENCES Apartments(ApartmentID)
52 );
53
54 -- Creating the Demographics Table
55 CREATE TABLE Demographics (
56     PopulationID INT PRIMARY KEY,
57     AreaID INT,
58     AgeGroup VARCHAR(10),
59     Female INT,
60     Male INT,
61     Year INT,
62     Quarter INT,
63     FOREIGN KEY (AreaID) REFERENCES AreaNames(AreaID)
64 );
```

7.3. Appendix C – SQL Code to Adjust Prices for Inflation

```
63
64 UPDATE Sales AS s
65 SET SqmPrice = s.Price / s.Size;
66
67 UPDATE Sales
68 SET AdjustedPrice = Price * i.inflation,
69     AdjustedSqmPrice = SqmPrice * i.inflation
70 FROM inflation i
71 WHERE EXTRACT(YEAR FROM Sales.SaleDate) = EXTRACT(YEAR FROM i.year);
72
```

Copenhagen Apartments

Market Overview Price Forecasts

This is an interactive dashboard for assessing the state and development of the apartment market in Copenhagen for the time period 2000-2023 (August).

Select city areas of interest to compare distribution, developments, median prices (inflation adjusted to 2022), and demographics.

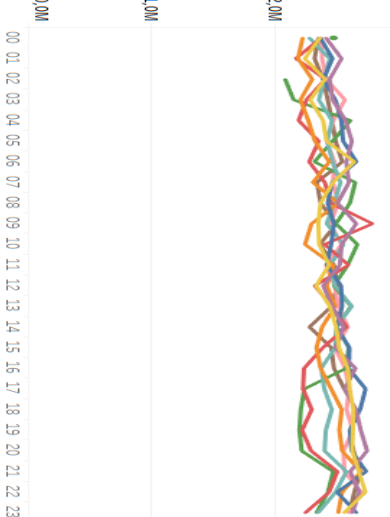
Below you can set the desired size and price range you want to examine. Switch between the market overview and forecasting prices with the labelled buttons above.



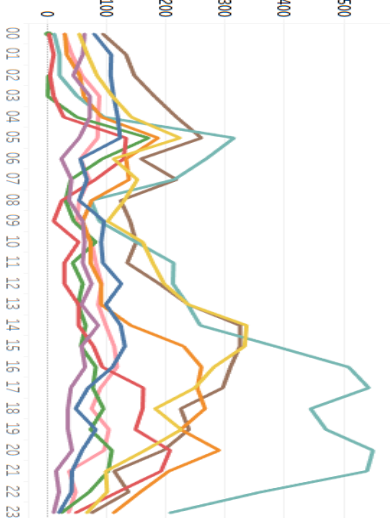
Apartment Size (sqm): Min. 0 Max. 1000 Select Area (All) ▼

Price Range: 2000 000 3 000 000

Price Development

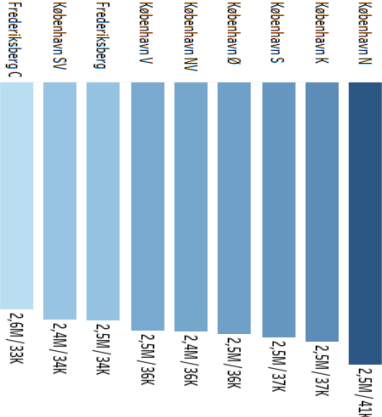


Sales Frequency

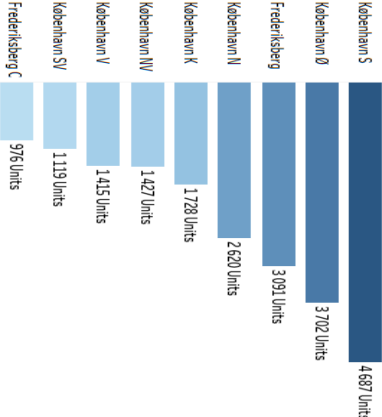


Filter on Price: True ▼ Filter on Size: True ▼ Exclude outliers: True ▼

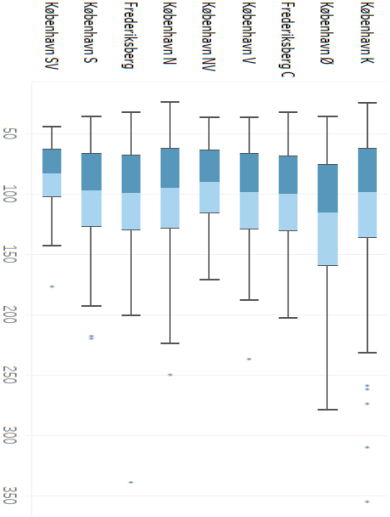
Apartment / Square Meter Price



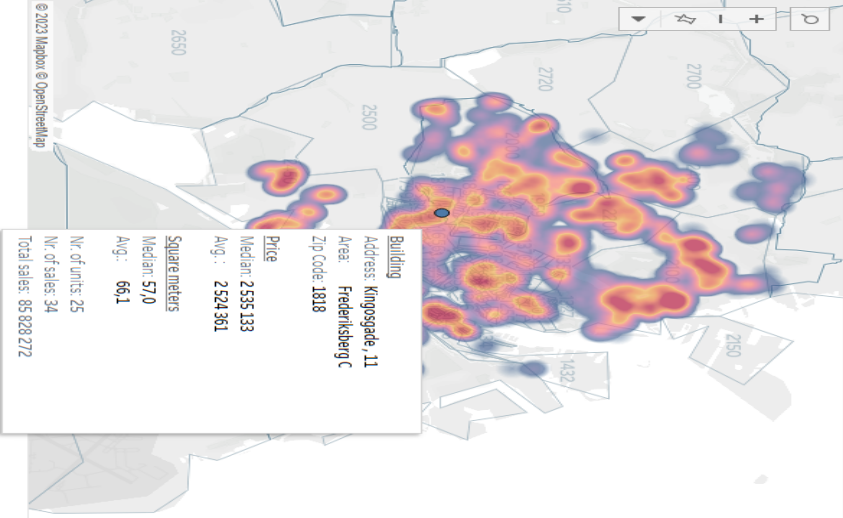
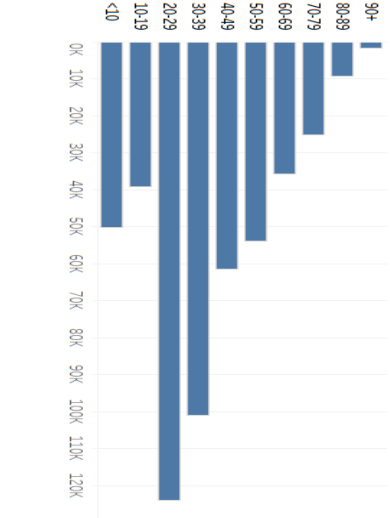
Apartments within Size & Price Range



Size Distribution



Age Distribution



Copenhagen Apartments

Market Overview

Price Forecasts

This is an interactive dashboard for assessing the state and development of the apartment market in Copenhagen for the time period 2000-2023 (August).

Below you can select the city area of interest and the desired size and price range (inflation adjusted to 2022) you want to forecast for. Switch between the market overview and forecasting prices with the labelled buttons above.

Apartment Size (\$QM):

Min.

0

Max.

1000

Select Area:

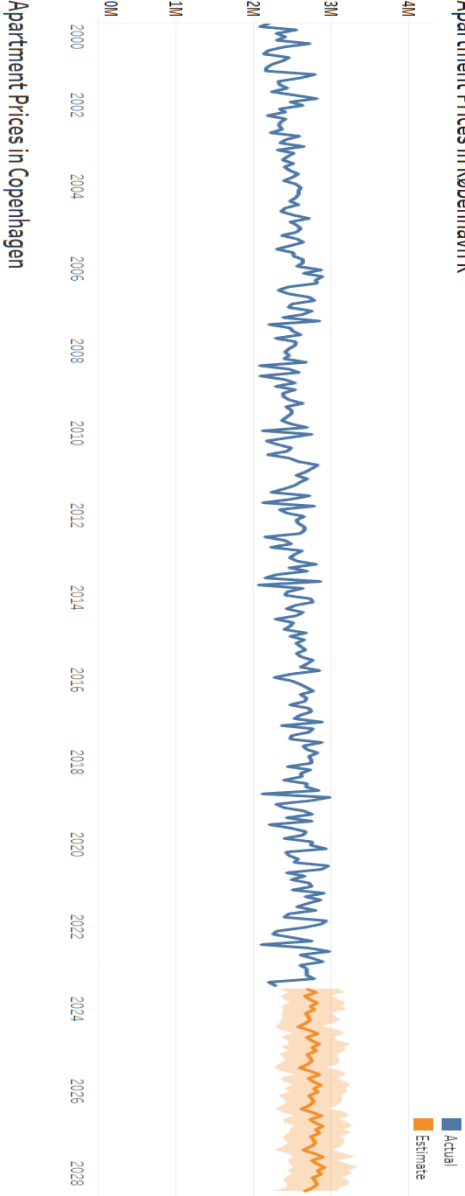
København K

Price Range:

2 000 000

3 000 000

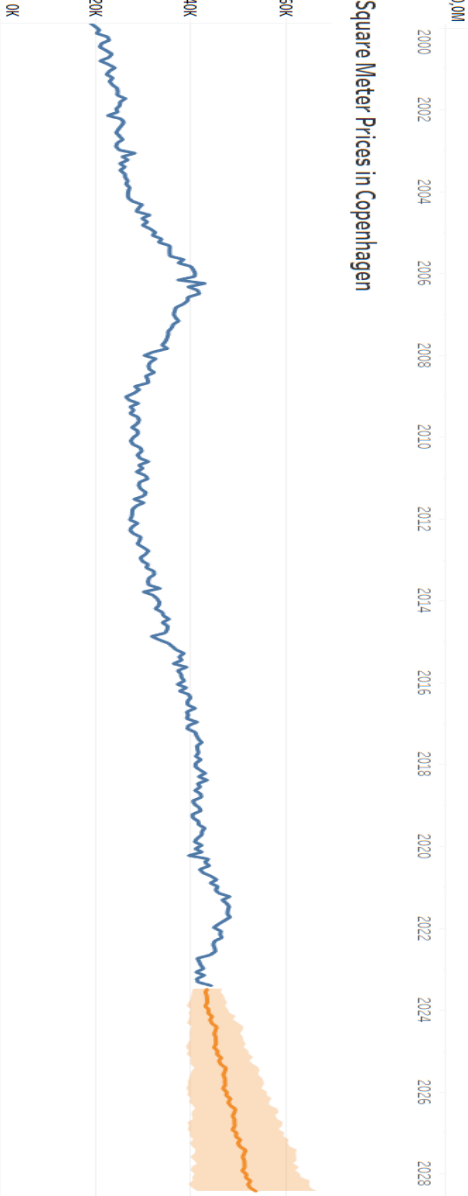
Apartment Prices in København K



Apartment Prices in Copenhagen



Square Meter Prices in Copenhagen



7.5. Appendix E – Dashboard: Detailed Picture of Price Forecasting